

Class 11 Biology: High-Yield Diagram & Mark Breakdown

Chapter 3: Plant Kingdom (NCERT Page-wise Priority Guide)

Instructions for Students: Practice distinguishing male and female reproductive thalli and sporophyte vs. gametophyte stages with clear labels.

Fig No.	Diagram / Organism	NCERT	Marks	Priority	Key Labelling Points / Concepts
Fig 3.2 (a, b)	<i>Marchantia</i> (Liverwort) Female & Male Thalli	Pg 28	3 M	Top Priority	Archegoniophore (female rays), Antheridiophore (male disc), Gemma cup, Thallus, Rhizoids
Fig 3.2 (c)	<i>Funaria</i> (Moss) Gametophyte & Sporophyte	Pg 28	3 M	Top Priority	Sporophyte: Capsule, Seta, Foot. Gametophyte: Leaves, Main axis, Rhizoids
Fig 3.1 (b)	Brown Algae (<i>Laminaria</i> , <i>Fucus</i> , <i>Dictyota</i>)	Pg 25	2 – 3 M	Top Priority	Frond (leaf-like), Stipe (stalk), Holdfast (attachment), Air bladder, Midrib (in <i>Fucus</i>)
Fig 3.3 (b)	<i>Equisetum</i> (Horsetail)	Pg 31	2 – 3 M	Top Priority	Strobilus (cone), Node, Internode, Branches, Underground Rhizome
Fig 3.3 (a, d)	<i>Selaginella</i> & <i>Salvinia</i>	Pg 31	2 – 3 M	High Priority	<i>Selaginella</i> : Stem, Microphylls, Roots, Strobilus. <i>Salvinia</i> : Floating aquatic fronds
Fig 3.1 (a, c)	Green & Red Algae (<i>Volvox</i> , <i>Polysiphonia</i>)	Pg 25	2 M	High Priority	<i>Volvox</i> : Daughter & Parent colony. <i>Polysiphonia</i> : Main axis, Branches. <i>Porphyra</i> : Frond
Fig 3.2 (d)	<i>Sphagnum</i> Gametophyte	Pg 28	2 M	High Priority	Antheridial branches, Archegonial branches, Branch clusters
Fig 3.4 (c)	<i>Ginkgo</i> (Gymnosperm)	Pg 33	2 M	High Priority	Long shoot, Dwarf shoot, Fan-shaped leaves, Naked seeds

EXAM STRATEGY & GUIDELINES

- **Liverwort Sex Organs:** In *Marchantia*, clearly distinguish female (umbrella/star-like rays) from male thallus (scalloped disc).
- **Funaria Structure (3 Marks):** Demarcate the dependent diploid sporophyte (Capsule, Seta) from the independent haploid gametophyte axis.
- **Table 3.1 (Pg 27):** Review pigment and food reserve combinations alongside algae diagrams.